

NETWORK SCANNER

UNDERSTANDING THE VULNERABILITIES WITH NMAP

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OBJECTIVE

- Main Goal: Utilize the Nmap tool to scan a device on a network, identifying open ports, detecting the operating system, and gathering other key information. This is a common practice for network administrators to map and secure their networks
- Scan network devices to discover open ports and running services.
- Identify operating systems and observe system behavior.
- Practice practical Nmap techniques (-sV, -O, -A) on Linux.
- Build real-world cybersecurity skills for professional development.

TOOLS USED

- VMware Workstation 17 Pro :

Desktop virtualization platform that runs multiple virtual machines (VMs) on one host for testing, sandboxing and OS-level experiments

- Kali Linux VM :

Debian-based distro packed with penetration-testing and forensics tools used for ethical hacking and security assessments.

- Nmap :

Fast network scanner for discovering hosts, open ports and running services. It's the go to recon tool for network mapping

STEP BY STEP PROCEDURE

- 1. Boot Kali Linux in VirtualBox/Vmware (e.g., Vmware Workstation 17 Pro).
- 2. Identify the target IP (use `ip a` / `ifconfig` or check the VM NAT gateway).
- 3. Run a basic port scan: `nmap <target_IP>` – note open ports.
- 4. Detect service versions: `nmap -sV <target_IP>` – record service names & versions.
- 5. Probe OS detection (if authorized): `nmap -O <target_IP>` – note likely OS/fingerprint.
- 6. Run a comprehensive scan (optional): `nmap -A <target_IP>` for combined OS, versions, scripts.
- 7. Save screenshots/logs of each scan and write a short analysis of key findings (open ports, outdated versions, unusual services).

COMMAND USED

Nmap 104.26.0.43

1. Purpose: Quick scan of the 1,000 most common TCP ports.
2. Benefit: Fast baseline of open ports and basic service hints.

COMMAND USED

```
Nmap -sV 104.26.0.43
```

1. Purpose: Probe open ports to identify service names and versions.
2. **Benefit:** Precise service info for CVE checks and targeted testing.

COMMAND USED

`Sudo nmap -O 104.26.0.43`

1. Purpose: OS/device fingerprinting using advanced TCP/IP probes.
2. Benefit: Guides OS-specific checks – but often low confidence behind proxies.

ANALYSIS AND RESULT

- Basic NMAP scanning
- Command : `nmap 104.26.0.43`

- **Analysis:**

Quickly scans the 1,000 most common TCP ports to find open services.

Gives a fast baseline of reachable services and potential attack surface.

May miss services on non-standard ports or UDP-only services.

Best used first to decide which ports or hosts deserve deeper probing.



```
sc05@KALI: ~  
File Actions Edit View Help  
(sc05@KALI)~  
$ nmap 104.26.0.43  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-09-29 21:13 IST  
Nmap scan report for 104.26.0.43  
Host is up (0.064s latency).  
Not shown: 996 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
8080/tcp   open  http-proxy  
8443/tcp   open  https-alt  
  
Nmap done: 1 IP address (1 host up) scanned in 7.04 seconds  
(sc05@KALI)~
```

ANALYSIS AND RESULT

- Service Version Detection scan
- Command: `nmap -sV 104.26.0.43`
- Analysis :

Probes open ports to fingerprint applications and report versions.

Provides actionable intel for vulnerability checks and targeted testing.

Can be confused by reverse proxies or CDN front-ends (shows proxy info).

Follow up with app-layer tests (HTTP headers, virtual hosts) if proxied.

```
(sc05@KALI)~[~]
$ nmap -sV 104.26.0.43
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-09-29 21:13 IST
Nmap scan report for 104.26.0.43
Host is up (0.057s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http         Cloudflare http proxy
443/tcp   open  ssl/https    cloudflare
8080/tcp   open  http         Cloudflare http proxy
8443/tcp   open  ssl/https-alt cloudflare

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.82 seconds
(sc05@KALI)~[~]
```

ANALYSIS AND RESULT

- Operating System Detection
- Command: `sudo nmap -O 104.26.0.43`
- Analysis :

Uses TCP/IP fingerprinting to guess the host OS - requires root for raw packets.

Useful when you need OS-specific exploits or configuration checks.

Often returns low-confidence or incorrect results behind load-balancers/CDNs.

If OS detection fails, pivot to application-layer and authenticated scans instead.

```
(sc05@KALI)-[~]
$ nmap -O 104.26.0.43
TCP/IP fingerprinting (for OS scan) requires root privileges.
QUITTING!

(sc05@KALI)-[~]
$ sudo nmap -O 104.26.0.43
[sudo] password for sc05:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-09-29 21:14 IST
Nmap scan report for 104.26.0.43
Host is up (0.050s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
8080/tcp   open  http-proxy
8443/tcp   open  https-alt
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.63 seconds
```

CONCLUSION

- Created a safe, isolated lab in VMware Workstation Pro (Kali Linux on a NAT network) to practice controlled network scanning.
- Performed Nmap scans (port discovery, `-sV` service/version detection, `-O` OS fingerprinting) to quickly map open ports, services, and host characteristics.
- Findings revealed actionable reconnaissance data and the limits of fingerprinting behind proxies/CDNs, highlighting when to pivot to application-layer or authenticated testing.
- The exercise strengthened practical reconnaissance skills and reinforced proper methodology, controlled environments, and ethical/authorized testing practices.

THANK YOU
